

**Social Science Majorship
Micro-Macroeconomics**

I. ECONOMICS

- *Oikonomia*: Greek word meaning “management of the household”
- Is a Social Science concerned with how individuals and society choose to use its scarce resources to achieve maximum satisfaction of human material wants.
- Economics is the social science that studies the production, distribution, and consumption of goods and services

Why study economics?

1. To learn a new way of thinking : Cost and benefit thinking
 1. **Scarcity**: is the natural limitation of resources that nature and previous generations have provided.
 2. **Opportunity cost**: the cost or value of the next best alternative or choice
 - That which we forgo or give up, when we make a choice or a decision.
 - “No such thing as free lunch”
 3. **marginalism**: in weighting the cost and benefit of a decision, it is important to weigh only the cost and benefit that arises from the decision.
2. To understand society
3. To understand global affairs
4. To become an informed voters

Tools in Economics: Theories and Models

Assumption: principle that economists use to simplify reality and make the world easier to understand. One important assumption used in economics is “*ceteris paribus*” or “*all other things being equal or constant*” it helps us simplify reality in order to focus on the relationship that we are interested in.

Variable: a measure that can change from time to time and from observation to observation. e.g. price, income, inflation, GDP, GNP

Theory: a statement or set of related statements about cause and effect, action and reaction.

Model: a formal statement of a theory. Usually a mathematical statement of a presumed relationship between two or more variables

DIVISION OF ECONOMICS

- **Microeconomics** - The branch of economics that examines the functioning of individual industries and the behavior of individual decision-making units, that is, business firms and households.
- **Macroeconomics** - The branch of economics that examines the economic behavior of aggregates – income, employment, output, and so on – in a national scale.

BASIC ECONOMIC QUESTIONS

- 1) **What to produce** – The economic system must decide what goods and services to produce with its land labor and capital
- 2) **How to produce** - The economic system must decide how to produce each good or service – determining what mix of land, labor, and capital to use in production methods to employ
- 3) **For whom to produce** – The economic system must decide which members of society will receive how much of the goods and services produced – the process of allocating income.

Economic Systems

1. **Market Economy** : This is also known as laissez-faire economy or free market economy. In this type of economy individuals and firms pursue their own self-interest without any central direction or regulation.
2. **Command Economy** : The basic economic questions are answered by a central government or planning agency. Through a combination of government ownership of enterprises and central planning the government dictates the directions of the economy.
3. **Mixed Economy** : A regulated market economy. The government plays a vital role in the economy while allowing the market mechanism to the govern economy.

Production: the process of making goods and services to satisfy human material wants and maximize profits.

Factors of Production

1. **Land**: includes all resources found on land, and on the sea.
2. **Labor**: it can be define as any kind of work, either mental or manual in nature, which has the sole purpose of receiving rewards. According to Alfred Marshall, labor can be define as any exertion that the mind or body undergone, either partly or totally, with the view of earning some other good other than the pleasure derived from work itself.
3. **Capital**: refers to the stock of goods made by the people to help them in the production of goods and services.
4. **Entrepreneurship**: coordinates all the factors of production to produce goods and services.

How the Market Works

Market: Is a mechanism by which the buyers and sellers interact to determine both price and quantity of goods and services

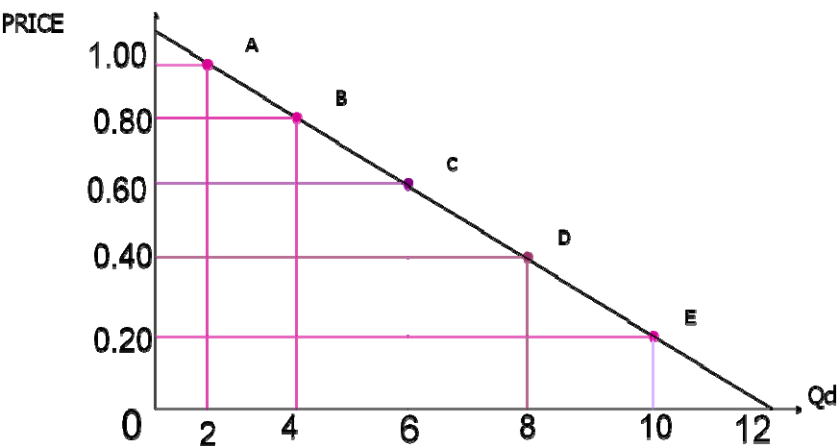
Demand and Supply and Price Determination

Demand: it is a market expression of the cumulative willingness and ability of household to buy different amount of product at different prices over a given period of time.

- **Demand schedule**: a table showing the different amount of a product that households will buy at all possible prices.

Price of Choc-nut	Quantities of Choc-nut Demanded
0	12
0.20	10
0.40	8
0.60	6
0.80	4
1.00	2

- Demand curve: a graphical representation of the demand schedule. It is a down-ward sloping curve.



- Law of Demand: Ceteris paribus, when price increase, quantity demanded increase, when price decrease, quantity demanded decrease. Relationship is inversely proportional or negative
- Demand Function: mathematical expression of the law of demand. $Qd = a - bP$

Exemptions to the law of Demand

- Giffen goods: a theoretical goods, when price increase demand also increases. (Sir Robert Giffin)
- Snob goods: when a lot of people buy the good, its demand died down.

Determinants of Demand

- Price: lead to a change in quantity demanded, causes a movement along the demand curve.
- Non-price: leads to a change in demand; cause a change in the whole demand schedule and demand curve

Factors of Demand

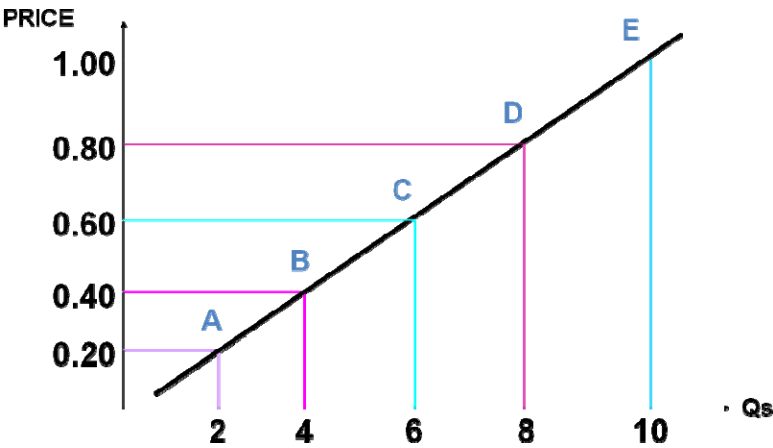
- Changes in income
- Tastes and Preferences:
- Price of substitute goods
- Price of complimentary goods
- Expectation of future income, wealth and prices
- Population
- Seasonality

Supply: Is a market expression of the cumulative willingness and ability of all firms to sell different amount of a product at different prices over a given period of time.

- Supply schedule: a table showing the different amount of a product that firm will sell at all possible prices over a given period of time.

Price of Choc-nut	Quantity Supplied of Choc-nut
0	0
0.20	2
0.40	4
0.60	6
0.80	8
1.00	10

- Supply curve: a graphical representation of the demand schedule. It is an up-ward sloping curve.



- Law of Supply: ceteris paribus, when price increase, quantity supply increase, when price decrease, quantity supply decrease.
- Supply Function: mathematical expression of the law of supply. $Q_s = a + bP$

Determinants of Supply

1. Price – lead to a change in quantity supplied, causes a movement along the supply curve
2. Non-price – lead to a change in supply, cause a shift in the supply curve

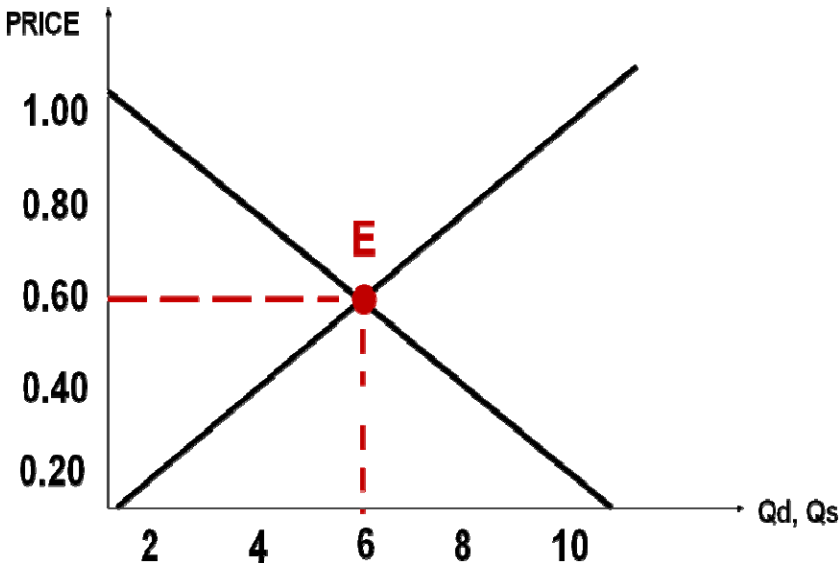
Factors of Supply

1. Cost of production
2. Price of related goods
3. No. of firms in the industry
4. Seasonality (weather condition)
5. Expectation of future price

Market Equilibrium

- The objective of the market is to reach a state in which the quantity of goods that producers are willing and able to supply and the quantity of goods that a buyers are willing and able to buy are equal at the same price
- Equilibrium : it refers to a state in which at the one and only price quantity supplied is equal to quantity demanded.

Price of Choc-nut	Quantity Demanded	Quantity Supplied
0	12	0
0.20	10	2
0.40	8	4
0.60	6	6
0.80	4	8
1.00	2	10



Disequilibrium Condition

1. Excess Demand : $Q_d > Q_s$ = shortage

- When quantity demanded exceeds quantity supplied, price tends to rise. When price rises, quantity demanded increases and quantity supplied decreases until equilibrium is reached.
- Excess Supply : $Q_d < Q_s$ = surplus / glut
 - When quantity supplied exceeds quantity demanded price tends to fall. When price falls, quantity supplied will likely decrease and quantity demanded will likely increase until equilibrium is reached.

Price Elasticity

- Measures the responsiveness of demand and supply, whenever price changes.
- Price Elasticity of Demand : The ratio of the percentage change in quantity demanded to the percentage change in price.
- $E = \frac{\% \text{ change in } Q_d}{\% \text{ change in } P}$

$$\% \text{ change in } Q_d = \frac{Q_{d2} - Q_{d1}}{Q_{d1} + Q_{d2}} \times 100$$

$$\% \text{ change in } P = \frac{P_2 - P_1}{P_1 + P_2} \times 100$$

Value of Elasticity Coefficient

Value	Type of Elasticity	Interpretation
1	Unitary	% change in Q_d = % change in P
> 1 (greater than 1)	Elastic	% change in Q_d > % change in P
< 1 (less than 1)	Inelastic	% change in Q_d < % change in P

Determinants
 of Price
 Elasticity of
 Demand

- nature of the good : If the good is luxury, more elastic. If the good is necessity,inelastic or less elastic
- Availability of substitutes : More substitute, more elastic, less substitute, less elastic
- share in the consumer’s income: If the share is small, inelastic or less elastic. If the share is big, more elastic
- adjustment time available : If the adjustment time is short, inelastic or less elastic. If the adjustment time is long, more elastic.

Market Regulation : Price Control

- Reasons for Price Control
 - The usual reason for the government to control the price of goods and services is “fairness”.
 - Emergency situations
- Types of Price control
 - Price Ceiling : people think that the market determined price is very high and the sellers are greedily amazing huge profit at their expense thus price is it always set below the market determined (equilibrium) price.
 - Price Floor : sellers think that the prevailing market determined price is too low for them to recover their production cost or earn a profit. Price is usually set at any price above the market price (equilibrium price).

Different Market Structures

- Perfect Competition : there are many small sellers in the market thus nobody control the price. Sellers are said to be price takers
- Monopoly : there is only one seller in the market; price is determined by the seller.
- Oligopoly : is a market where there are few big sellers colluding with each other.
- Monopolistic competition : involves a few sellers supplying the market with different types of brands of the same product and capturing a specific group of buyers for a particular brand of product.
- Monopsony – the reverse of a monopoly, there is only one buyer for a particular good or service.

Macroeconomic Studies

Circular Flow Diagram of Economic Activities : a visual model of how the economy works as a whole

Measuring Growth : National Income Account

- Gross National Product (GNP)
 - The total market value of all the goods and services produced by a nation in a given period
- Gross Domestic Product (GDP)
 - Measures the total market value of all the goods and services produced in the country

Approaches to Measuring GDP

- Expenditures approach – adding up all the value of all the final goods and services spend during a given period.
 - Private consumption (C)
 - Government consumption (G)
 - Gross Private Investment (I)
 - Net Export (x-m)
 - $C+I+G+(x-m)$
- Income approach – measures the total income earned by the factors of production.

3. Production approach/ Value Added approach – adding up the contribution of the 3 major sectors of the economy; agriculture, industry, service

Macroeconomic Concerns

- **Business Cycle** : The cycle of short-term ups and downs of in the economy
- **Recession** : a period during which aggregate output declines for two consecutive quarter
- **Depression**: a prolonged and deep recession
- **Inflation** : Is a sustained and general increase in prices in all or nearly all the markets in the economy.
- **Deflation** : a sustained and general decrease in prices in all of nearly all the markets in the economy

Types of Inflation

1. Demand-pull inflation – state of excess aggregate demand.
2. Cost push inflation – cause by a significant and continued increase in some cost of production (labor, raw materials, profit)

Effects of Inflation

1. Decrease in the value of money (Purchasing Power of the Peso)
2. Inflation loser
 - Fixed salary worker and retirees living on pension
 - Creditors and savers
3. Inflation gainers
 - Speculators
 - People with flexible income
 - Debtors

Government Role in Combating Inflation

- **Fiscal Policy** – is the manipulation of government spending and imposition of taxes
 - a. *Expansionary fiscal policy* – this commonly known as “pump priming” the economy characterized by a decrease in tax rate and increase in government spending to spur growth and employment
 - b. *Contractionary fiscal policy* – increase in taxes and decrease in government spending to combat inflationary pressures in the economy.
- **Monetary Policy** – manipulation of savings and investment of the financial sector to achieve the twin objectives of the macroeconomy
 - a. *Easy money policy* – increase in the supply of money (MS) in the economy to spur growth and increase economic activity
 - b. *Tight money policy* - decrease in the supply of money (MS) in the economy to fight inflationary pressure

D.) International Trade

- **Trade surplus**: when a country's export is more than its import.
- **Trade deficit**: when a country's import is more than its export.

The Economic Basis for Trade

- a) **Theory of Absolute Advantage**: an advantage in the production of a product enjoyed by one country over another when it uses fewer resources to produce that product than the other country.
- b) **Theory of Comparative Advantage**: The advantage in the production of a product enjoyed by one country over another when that product can be produced at a lower cost in terms of the other goods than it could be in the other country.

Exchange Rate: the ratio at which two currencies are traded. The price of one currency in terms of another.

- **Floating or Market determined exchange rate**: exchange rates that are determined by the unregulated forces of supply and demand.
- **Appreciation of currency**: the rise in value of one currency over the other
- **Depreciation of currency**: the fall in value of one currency over the other

Economic integration: occurs when two or more nations join to form a free-trade zone. e.g. European Union (EU), Asia-Pacific Economic Cooperation (APEC), World Trade Organization (WTO).

III. ECONOMIC PLANNING AND DEVELOPMENT

A.) Economic Planning

- A centralized national planning is important because it provides an institutional mechanism to identify national economic problems and coordinate efforts to solve or minimize these problems.
- The responsibility for Economic Planning was vested in the National Economic and Development Authority (NEDA). Created in January, 1976 with the mandate for both macroeconomic planning and project planning and implementation.

Reasons why development planning is important to developing nations.

Market Failure

The market of developing countries is not perfect and is full of factors and unexpected instances that may affect the flow of economic elements. Environmental pollution is a factor that could affect the market and the consequently, the economy. The local residents are usually affected by the effects of pollution. This only means that the allocation of products and services in the society is not effective, and the public interest is not protected. Planning is important for the government to come up with ways to resolve these problems between consumers and investors.

Resource Mobilization and Allocation

Developing countries must use its limited financial resources and human resources sparingly and wisely. Public sector projects need to be selected meticulously so limited resources do not go to waste.

Attitudinal and Psychological Impact

Developmental plans outline the government objectives that will satisfy the needs and requirements of its citizenry. Developmental plans greatly impact on the psyche of a people as these are what will give direction to a nation's road to progress. A clear realistic and well-planned set of programs shall no doubt earn public support and elicit cooperation. Without such support, a program may not prosper.

Foreign Aid

Having a detailed plan that specifies the priority public works is an important condition for the country to obtain foreign grants and assistance. It is easier to convince foreign donors and investors to invest when the plans are detailed and laid out properly.

B. Development

1.) Etymology and Definition

- "Economic development is the upward movement of society towards the direction of modernization ideals. These ideals are as follows: rationality planning for development, increase in productivity, increase in social and economic equality, increase in standard of living, increase in social and economic equality, more efficient institution and attitude conducive to increased productivity and development consolidation of the national state and national integrity, national independence, political democratization and social discipline " Gunnar Myrdal, Asian Drama: An inquiry into the Poverty of Nations" (1968).
- Development...as a multidimensional process involving the reorganization and reorientation of entire economic and social system. Michael P. Todaro, Economic Development in the Third World, (Fourth Edition)

2.) Development Economics:

- Common Characteristics of Developing Nation (Michael Todaro)
 - 1.) low levels of living
 - 2.) low level of productivity
 - 3.) high rates of population growth and dependency burdens
 - 4.) High and rising levels of unemployment and underemployment
 - 5.) Significant dependence on agricultural production and primary product
 - 6.) Dominance, dependence and vulnerability in international relations

3.) Measures of Development

- The most popular indicator of economic development are per capita Gross National Product (GNP) and per capita Gross Domestic Product (GDP)
- Other indicators of development
 1. **Physical Quality of Life Index (PQLI)**: is an attempt to measure the quality of life or well-being of a country. The value is the average of three statistics: basic literacy rate, infant mortality, and life expectancy at age one, all equally weighted on a 0 to 100 scale.
 2. **Human Development Index (HDI)**: The HDI combines normalized measures of life expectancy, literacy, educational attainment, and GDP per capita for countries worldwide. It is claimed as a standard means of measuring human development—a concept that, according to the United Nations Development Program (UNDP), refers to the process of widening the options of persons, giving them greater opportunities for education, health care, income, employment, etc. The basic use of HDI is to measure a country's development.
 3. **Capability Poverty Measure (CPM)**: The United Nation Development Program (UNPD) first introduced the CPM in 1996. The CPM considers three objectives of development namely; to be nourished and healthy, to be capable of safe and healthy reproduction, and to be knowledgeable and educated. Thus the indicators are longevity, reproductive health and knowledge.
 4. **Economic Freedom Index**: The Heritage Foundation Inc. regularly constructs the economic freedom index. It uses ten criteria: trade, taxation, government intervention, monetary policy, foreign investment, banking, wage and price, property rights, regulation, and black market activities.

4.) Leading Theories of Economic Development

A. Growth Stage Theories

- Growth stage theories represent an effort to systematize the process of economic growth within a framework of a sequence of stages applicable across national and cultural boundaries.
 - a.) List and the German Historical School
 - List based his growth stage classification on shifts on occupational distribution. According to List the five stages of growth are as follows;
 - (i) Savage, (ii) Pastoral, (iii) Agriculture, (iv) Agricultural Manufacturing, (v) Agricultural-Manufacturing Commercial.
 - b.) Karl Marx
 - Marx's criteria of stage classification are type of technology used, system of property rights and ideology behind the technology. The stages of growth according to Marx are as follow; (i) Primitive communist (communal formation), (ii) chattel/ancient slavery, (iii) stages of empire, (iv) medieval feudalism, (v) industrial capitalism, (vi) industrial capitalism, (vii) pure communism-socialism.
 - c.) Walt Whitman Rostow ("Leading Sector" Stage Theory)
 - In 1960, he wrote his famous book, "The Stages of Economic Growth: A Non-Communist Manifesto".
 - According to Rostow, all developing countries have linear progressed from transitional economies to modern economies. They all have gone through the following stages; (i) traditional stages, (ii) precondition to take-off, (iv) take-off into self-sustaining growth, (v) drive to maturity, (v) age of high mass consumption

C. Dependency Theory

- The coexistence of rich and poor nations in an international system dominated by unequal power relationship between the center (the developed countries) and the *periphery* (the LDCs) renders attempts by poor nations to be self-reliant and independent in their development efforts difficult and sometimes even impossible.

D. Neoclassical Counter-Revolution

- The central argument of the neoclassical counter-revolution is that underdevelopment results from poor resource allocation due to incorrect pricing policies and to too much state intervention by overly active Third World government.

5.) Issues in Economic Development

a) Poverty and Income Distribution

Poverty is a condition in which a person or community is deprived of, or lacks the essentials for a minimum standard of well-being and life.

A **poverty line** or **poverty threshold** is a benchmark that that helps determines the levels of poverty by drawing a line separating those who are poor (those below the poverty line) and the non-poor (those above the poverty line). The Philippine government uses the Family Income and Expenditure Survey (FIES) a survey conducted by the National Economic and Development Authority (NEDA) to establish the poverty threshold.

Region/Province	Annual Per Capita Poverty Threshold (in Pesos) d/				Magnitude of Poor Families		
	1991 a/	2003	2006	2009	Estimates		
					2003	2006	2009
PHILIPPINES	5,776	10,976	13,348	16,841	3,293,096	3,670,791	3,855,730

Source: NEDA

Income Distribution

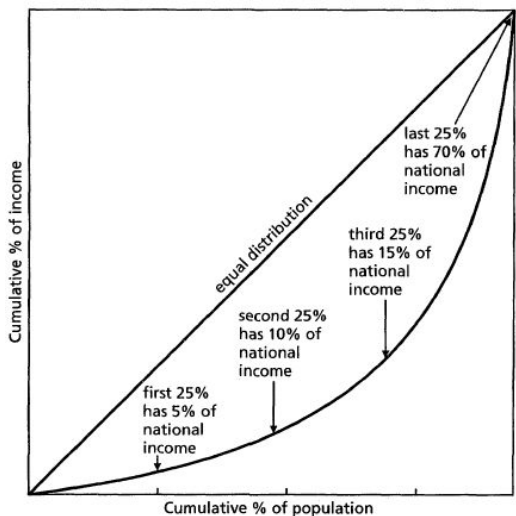
Income distribution is an integral part of the issues of poverty. There are two distinct ways by which income distribution might be discussed.

- 1) **Size distribution of income classes:** the approach used when the family incomes and expenditures were arranged by income classes. This underlies the use of poverty line as a measure of poverty.
- 2) **Functional Income Distribution:** how factors of production, through their distinct contribution to the production process, derive their share of output. This calls for analyzing factors shares in the total income within the country.

Lorenz Curve and Gini coefficient

The Lorenz curve is graphical representation of wealth distribution developed by American economist Max Lorenz in 1905. On the graph, a straight diagonal line represents perfect equality of wealth distribution; the Lorenz curve lies beneath it, showing the reality of wealth distribution. The difference between the straight line and the curved line is the amount of inequality of wealth distribution, a figure described by the Gini coefficient.

A diagonal line shows an even distribution, and the calculation of the Gini coefficient uses the 'gap' between the diagonal and the actual curve. The coefficient, also known as *Gini's concentration ratio*, may be calculated as the ratio of area between the diagonal and the Lorenz curve to the total area beneath the diagonal. The lower the Gini coefficient, the more evenly spread the variable.



Kuznet's Hypothesis

Simon Kuznets, the economist who had devoted much of his professional life to an analysis of economic data of nations. He observed that income distribution was once relatively equal before rapid growth took place. Once rapid growth set in, the income distribution worsened for a prolonged period. But later, income distribution improves again as the countries concerned devised social policies that promote greater equalization of incomes and opportunities. This is known as *Kuznet's Hypothesis*.

b) Labor and Employment

- **Labor force:** the members of the population who are 15 years old and above and who are holding a job or are seeking employment for productive work represents the labor force
- **Labor Force Participation Rate:** the percentage of the population belonging to the labor force or is the ratio between the labor force and the total population 15 years old and over.
- **Employment Rate:** the percentage of the labor force that is employed.
- **Unemployment Rate :** the percentage of the labor force that that desires to be employed but cannot find a job.

Types of Unemployment

- 1) **Frictional unemployment:** people who are not working temporarily because they are in between jobs. It can never be zero (0)
- 2) **Cyclical unemployment:** brought about by the operation of the business cycles. After an upsurge in output comes overproduction and therefore firms begin to count on their inventories and start reducing their work force.
- 3) **Structural unemployment:** it is similar to frictional unemployment but tend to last for years. It arises from mismatch of skills needed versus what is available.

Underemployment Rate: the percentage of the labor force who are working but who do not render the full equivalent of working time either by choice or by inability to find full-time work.

Self-employment: who renders their labor to contribute to their own or family undertakings. Some economists call this employment as disguise unemployment

Unemployment and underemployment are the twin phenomenons that co-exist in developing countries like the Philippines. Normally, unemployment is a situation which some people willing to work for income are not able to do so because they cannot find employment opportunities. But in developing countries like the Philippines, those normally considered as unemployed are partly engage in self-employed activities engage in the informal sector, mostly in retail.

Productivity and Wages

Productivity: the amount of goods and services produced from each hour of the workers time.

Unit labor cost:

- Measures the cost of one unit of labor when hired by an employer. The numbers may be measure initially as per day cost. When compared internationally, the cost are converted into dollars and then compared as index number. It is a measure of competitiveness when compared with other counties or other industries.
- All factors that enter the wage rate are reflected in the unit cost of labor: minimum wage, productivity of labor, cost of union activity, the efficiency of infrastructure in supporting the business environment and so on.

Minimum wage

- A legislated floor price for wage rates – a minimum hourly rate for any kind of labor.
- The government discarded the system of national minimum wage system upon the passage of Republic Act 6727 which created the *Regional Tripartite Wages and Productivity Boards* to rule on the determination of regional wage rate setting. The membership of the Board includes the representative from labor and employers associations.

Other issues on labor

- Militant unionism
- Contractual casual employment scheme
- Labor services subcontracting scheme

